



Islamic University of Technology
Department of Computer Science and Engineering

Lab 10: Greedy Algorithms

CSE 4404: Algorithms Lab(SWE)
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Contents

Task A. Fractional Knapsack	2
Task Statement	2
Input Format	2
Output Format	2
Constraints	2
Examples	2
Task B. GPU Cluster Scheduling	3
Task Statement	3
Input Format	3
Output Format	3
Constraints	3
Examples	3
Task C. Fruit Ninja Champion	4
Task Statement	4
Input Format	4
Output Format	4
Constraints	4
Examples	4

Task A. Fractional Knapsack

Time Limit: 1 second | Memory Limit: 256 MB

You are a thief with a knapsack of capacity W . There are N items available, where item i has weight w_i and value v_i . Unlike the classic 0/1 knapsack, you are allowed to take any **fraction** of an item: taking a fraction x (where $0 \leq x \leq 1$) of item i costs $x \cdot w_i$ capacity and earns $x \cdot v_i$ value. Determine the maximum total value you can carry without exceeding the knapsack's capacity.

Since the maximum achievable value may not be an integer, print the **floor** of the maximum total value.

Input Format

The first line contains two integers N and W . The next N lines each contain two integers: w_i and v_i , the weight and value of item i .

Output Format

Print a single integer — the floor of the maximum total value that can be carried in the knapsack.

Constraints

- $1 \leq N \leq 10^5$
- $1 \leq W \leq 10^9$
- $1 \leq w_i, v_i \leq 10^9$

Examples

Sample Input	Sample Output
3 50 10 60 20 100 30 120	240
1 5 10 60	30

Task B. GPU Cluster Scheduling

Time Limit: 1 second | Memory Limit: 1536 MB

Imagine you manage a computer system where engineers run AI training jobs on shared graphics cards (GPU clusters). Each job i runs during a specific time window, where s_i is the start time and f_i is the finish time. The exact length or duration of each task is mathematically defined as: $d_i = f_i - s_i$. Because two jobs cannot use the same GPU at the same time, your goal is to find the fewest GPUs needed to run them all. By sorting the jobs by their start times and automatically reusing a GPU the exact moment a previous job's duration d_i is complete, you can efficiently schedule the entire workload with the minimum amount of hardware.

Input Format

Each test case starts with a single integer N , representing the total number of jobs. The next N lines each contain two integers: s_i and f_i , representing the start time and finish time of job i .

Output Format

For each test case, print a single integer representing the minimum number of GPUs required.

Constraints

- $1 \leq N \leq 10^5$
- $0 \leq s_i < f_i \leq 10^9$

Examples

Sample Input	Sample Output
3 1 5 1 5 1 5	3
5 1 3 2 4 3 5 4 6 5 7	2

Task C. Fruit Ninja Champion

Time Limit: 2 seconds | Memory Limit: 512 MB

The screen is overflowing with fruits, each ready to fall if you don't slice them in time. Each fruit i has a fall time f_i and a point value p_i you earn if you slice it before it falls. You can slice at most one fruit per unit of time, and must slice a fruit at some integer time $t \leq f_i$. Your mission is to slice fruits smartly to maximize your total points and become the ultimate Fruit Ninja Champion!

Input Format

The first line contains an integer T — the number of test cases. Each test case consists of:

- A line with an integer n — the number of fruits.
- A line with n integers $f_1, f_2, \dots, f_n$ — the fall times of the fruits.
- A line with n integers $p_1, p_2, \dots, p_n$ — the points for each fruit.

It is guaranteed that the sum of n over all test cases does not exceed 2×10^5 .

Output Format

For each test case, print a single integer — the maximum total points you can earn.

Constraints

- $1 \leq T \leq 10$
- $1 \leq n \leq 10^5$
- $1 \leq f_i \leq 10^9$
- $1 \leq p_i \leq 10^9$
- $\sum n \leq 2 \times 10^5$

Examples

Sample Input	Sample Output
1 3 2 1 2 10 20 30	50

Marks Distribution

Task	Marks
Task A	30%
Task B	35%
Task C	35%

Practice Problems

Problem	Link
Movie Festival	CSES: https://cses.fi/problemset/task/1629
Movie Festival II	CSES: https://cses.fi/problemset/task/1632
Tasks and Deadlines	CSES: https://cses.fi/problemset/task/1630